



Restaurant Sales Data Analysis

A Pandas Data Cleaning & Business Analysis Project

Dataset — 5,000 Transactions

File: restaurant_sales_data.csv

Role: Data Analyst — Restaurant Company



ALPHA MODEL



FACULTY OF AI

Project Overview

You are a Data Analyst working for a restaurant company. Your manager has handed you a raw sales export that is intentionally messy — the kind of file that shows up in real business life. Your job is to clean it, analyze it with Pandas, and turn it into insights the business can act on.

This project applies every Pandas technique covered from the start of the course through Session 8. Work through the phases in order — cleaning always comes before analysis.

The Dataset

You will work with `restaurant_sales_data.csv` — a dirtied, 5,000-row sample built specifically for this assignment from a larger 17,534-row source dataset. Every issue listed below was injected on purpose, so don't assume anything is clean until you've checked it yourself.

Column Reference

Column	Description	Example
Order ID	Unique identifier for each order.	ORD_123456
Customer ID	Unique identifier for each customer.	CUST_001
Category	Category of the purchased item. May be inconsistently capitalized/spaced.	Main Dishes
Item	Name of the purchased item. May be missing.	Grilled Chicken
Price	Price of the item. May be missing, negative, or stored as text (e.g. with \$).	15.0 / 15\$
Quantity	Quantity ordered. May be missing, negative, or stored as text.	3 / "three"
Order Total	Price × Quantity. May be missing.	45.0
Order Date	Date the order was placed. May contain invalid strings.	2022-01-15
Payment Method	Payment method used. May be missing or inconsistently formatted.	Cash / CASH

Known Data Issues (what to hunt for)

The table below is a starting map, not a complete answer key — go verify each of these yourself with `isna()`, `duplicated()`, `dtypes`, and `value_counts()`.

Issue Type	Affected Column(s)	Example Values Found
Missing values	Item, Price, Quantity, Order Total, Payment Method	empty cell / NaN
Duplicate rows	entire row	10 exact duplicate transactions appended
Wrong data type — text in numbers	Quantity	"two", "three", "4 items", "3 items"

Issue Type	Affected Column(s)	Example Values Found
Wrong data type — text in numbers	Price	"6.0\$", "15\$", "10.0\$"
Invalid numeric values	Quantity, Price	negative numbers, e.g. -2, -5.0
Inconsistent capitalization / spacing	Payment Method	"Cash", "cash", "CASH", "Cash Payment", " Cash", "e-wallet"
Inconsistent capitalization / spacing	Category	"Main Dishes", "main dishes", "MAIN DISHES", "Main Dishes"
Invalid / broken dates	Order Date	"abc", "2023/99/99", "00/00/0000", "31-02-2023", "2022-13-05"

Menu Reference (ground truth prices)

Use this table to sanity-check or reconstruct Price / Order Total when values are missing or broken.

Category	Item	Price (\$)
Starters	Chicken Melt	8.0
Starters	French Fries	4.0
Starters	Cheese Fries	5.0
Starters	Sweet Potato Fries	5.0
Starters	Beef Chili	7.0
Starters	Nachos Grande	10.0
Main Dishes	Grilled Chicken	15.0
Main Dishes	Steak	20.0
Main Dishes	Pasta Alfredo	12.0
Main Dishes	Salmon	18.0
Main Dishes	Vegetarian Platter	14.0
Desserts	Chocolate Cake	6.0
Desserts	Ice Cream	5.0
Desserts	Fruit Salad	4.0
Desserts	Cheesecake	7.0
Desserts	Brownie	6.0
Drinks	Coca Cola	2.5
Drinks	Orange Juice	3.0
Drinks	Lemonade	3.0
Drinks	Iced Tea	2.5
Drinks	Water	1.0
Side Dishes	Mashed Potatoes	4.0
Side Dishes	Grilled Vegetables	5.0
Side Dishes	Side Salad	3.0
Side Dishes	Garlic Bread	4.0
Side Dishes	Onion Rings	5.0

Project Rules

- Use Python and Pandas only.
- Do not use Matplotlib or Seaborn for this project.
- Organize your notebook or script according to the phases below.
- Do not skip the cleaning process — dirty data cannot be analyzed directly.
- Briefly explain every important cleaning and analysis decision.
- Answer all business questions using results from your own analysis.
- Do not just print results — explain what they mean for the restaurant business.

Phase 1 — Data Loading & Inspection

Required Tasks

1. Import Pandas.
2. Load the dataset.
3. Display the first 10 rows.
4. Display the last 5 rows.
5. Check the shape of the dataset.
6. Display the column names.
7. Check data types using dtypes.
8. Use info().
9. Use describe().
10. Check unique values in important categorical columns.

Questions to Answer

- *How many rows and columns does the dataset contain?*
- *Which columns contain missing values?*
- *Which columns are numerical? Which are categorical?*
- *Are any columns using an incorrect data type?*

Phase 2 — Data Cleaning

2.1 Missing Values

11. Check the number of missing values in each column.
12. Identify the columns with the highest number of missing values.
13. Decide how each missing value should be handled.
14. Use appropriate techniques: isna(), isnull(), fillna(), dropna().

Business Logic

Use the information you do have to reconstruct what's missing:

- If Price is missing but Order Total and Quantity are available, calculate Price.

- If Quantity is missing but Order Total and Price are available, calculate Quantity.
- If Order Total is missing but Price and Quantity are available, calculate Order Total.
- If a value can't be reliably reconstructed, choose an appropriate method and explain why.

Questions to Answer

- Which column has the most missing values?
- How many missing values remain after cleaning?
- Which method did you use for each important missing-value problem, and why?

2.2 Duplicates

15. Check duplicated rows using `duplicated()`.
16. Display the duplicated records.
17. Remove inappropriate duplicates using `drop_duplicates()`.
18. Verify that duplicates were removed.

Question to Answer

- How many duplicate records were found and removed?

2.3 Wrong Data Types

Inspect the data types and identify columns that should have a different type — for example Price, Quantity, Order Total, or Order Date. This dataset deliberately mixes text into numeric columns (like "two", "4 items", or "15\$"), so a simple `astype()` will fail until you clean the text first.

Useful techniques: `astype()`, `pd.to_numeric()`, `pd.to_datetime()`. Handle invalid values appropriately (`errors='coerce'` is your friend, but explain what you do with the NaNs it creates).

Questions to Answer

- Which columns had incorrect data types?
- What should their final data types be?
- Were there invalid values that prevented conversion?

2.4 Inconsistent Values

Check categorical columns — especially Category and Payment Method — for different capitalization, different spellings, extra spaces, and different representations of the same value.

Useful techniques: `replace()`, string methods (`.str.strip()`, `.str.title()`), `map()`.

Questions to Answer

- Which columns contained inconsistent values?
- What inconsistencies did you find?
- How did you standardize them?

2.5 Data Validation

After cleaning, re-check everything:

19. Check missing values again.
20. Check duplicates again.

21. Check data types again.
22. Check for invalid numerical values (e.g. negative Price or Quantity).
23. Verify that calculated values are logically consistent ($\text{Order Total} = \text{Price} \times \text{Quantity}$).

Final Cleaning Question

- *Is the dataset ready for analysis? Explain why.*

Phase 3 — Data Selection & Filtering

Work from your cleaned dataset for the rest of the project.

3.1 Selecting Columns

Use `[]` to select one column, then multiple columns. Explain the difference between the two.

3.2 `loc[]`

Perform at least 2 tasks using `loc[]` to select specific rows and/or columns.

3.3 `iloc[]`

Perform at least 2 tasks using `iloc[]` to select data by position.

3.4 Boolean Filtering

Perform at least 3 business-related filtering tasks — e.g. orders above a certain value, orders with quantity above a certain value, orders from a specific category.

3.5 `query()`

Use `query()` for at least 2 business-related filtering tasks.

3.6 Sorting

Use `sort_values()` for at least one ascending sort and one descending sort.

Questions to Answer

- *Which orders have the highest order totals?*
- *Which items have the highest quantities?*
- *What does the sorted result reveal about restaurant sales?*

Phase 4 — Data Transformation

4.1 Adding New Columns

Create at least 2 meaningful new columns — for example `Order_Value_Category`, `Order_Size`, `Revenue_Per_Item`, or `Month`. Explain why each one is useful.

4.2 `apply()`

Use `apply()` for at least one business-related transformation.

4.3 `lambda`

Use a lambda function for at least one transformation.

4.4 `map()`

Use `map()` for at least one meaningful transformation.

Questions to Answer

- *What new columns did you create, and why are they useful?*
- *What business problem does each transformation help solve?*

Phase 5 — Data Analysis

5.1 groupby()

Perform at least 4 groupby analyses using relevant dimensions such as Category, Item, Payment Method, Customer, or Month.

5.2 agg()

Use agg() to calculate multiple metrics at once — total revenue, average order value, total quantity sold, number of orders, average price. Build at least 2 aggregated summary tables.

Phase 6 — Business Questions

For every question below: write the Pandas code, display the result, and explain what the result means for the business.

Sales Performance

24. Which category generates the highest total revenue?
25. Which item generates the highest total revenue?
26. Which item has the highest quantity sold?
27. What is the average order value?

Customer Analysis

28. Which customers generate the highest total spending? Identify the top 5.
29. How many orders does the average customer make?

Payment Analysis

30. Which payment method is used most frequently?
31. Which payment method generates the highest revenue?

Category Analysis

32. Which category has the highest average order value?
33. Which category has the highest quantity sold?

Product Performance

34. What are the top 5 best-selling items by quantity?
35. Which items generate high revenue but relatively low sales volume? What might this indicate about pricing or demand?

Time Analysis

36. Which month generates the highest revenue?
37. Which month has the highest number of orders?

Business Performance

38. Which category or item performs below the others? Explain why you consider it a weak performer.
39. Are there any items with high quantity sold but relatively low revenue? What could this mean?

Phase 7 — Final Business Insights

Write at least 5 meaningful business insights. Don't just list numbers — use this structure for each one:

Finding → Evidence → Business Meaning

Required Insight Questions

40. What is the strongest-performing category? Why?
41. What is the strongest-performing item? Why?
42. Which payment method is most important for the restaurant? Why?
43. Which customers contribute the most revenue?
44. Which category or item requires improvement? Why?
45. What purchasing patterns can be observed?
46. What business actions would you recommend based on the results?

Final Conclusion

Write a short conclusion containing:

- The most important finding.
- The strongest category / item.
- The weakest category / item.
- One important customer-related finding.
- At least 2 practical business recommendations.

Final Deliverables

Submit:

47. Python notebook (.ipynb) or Python script (.py).
48. The cleaned DataFrame.
49. Code covering all required phases.
50. Relevant analysis tables.
51. Answers to the business questions.
52. At least 5 business insights.
53. Final business recommendations.

Skills Checklist

Before submitting, make sure your notebook uses every item below.

☐ Pandas import & setup	☐ read_csv()
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☐ head() / tail()	☐ shape
☐ columns	☐ dtypes
☐ info()	☐ describe()
☐ isnull() / isna()	☐ fillna()
☐ dropna()	☐ duplicated()
☐ drop_duplicates()	☐ replace()
☐ astype()	☐ pd.to_numeric()
☐ pd.to_datetime()	☐ Column selection ([])
☐ loc[]	☐ iloc[]
☐ Boolean filtering	☐ query()
☐ sort_values()	☐ Adding new columns
☐ apply()	☐ lambda functions
☐ map()	☐ groupby()
☐ agg()	☐ Business analysis & insights
☐ Final recommendations	☐

Important — this project is about understanding the data and solving a business problem, not just running Pandas functions.

Your code should answer: What happened?

Your analysis should answer: Why is it important?

Your final insights should answer: What should the business do next?